

4(a)	Simple harmonic motion is defined as an oscillatory motion in which the <u>acceleration</u> of an object is <u>directly proportional to the displacement</u> of the object <u>from its equilibrium position</u> , and the acceleration is always <u>directed towards that position</u> .	[1]
4(b)(i)	Angular frequency $\omega = \frac{2\pi}{T} = \sqrt{\frac{k}{m}}$ $= \sqrt{\frac{18.5}{0.400}} = 6.80 \text{ rad s}^{-1}$ Amplitude $x_0 = \frac{v_{\max}}{\omega} = \frac{1.40}{6.80} = 0.206 \text{ m}$	[1] value for ω [1] answer
4(b)(ii)	Maximum tension of the spring occurs at the lowest position. At the lowest position, acceleration is maximum $= (6.80)^2(0.206)$ $= 9.53 \text{ m s}^{-2}$ At lowest point, $T - mg = ma$ $T = mg + ma$ $= (0.400)(9.81) + (0.400)(9.53)$ $T = 7.74 \text{ N}$	[1] value for max acceleration [1] sub [1] ans

No.	Solution	Remarks
4(c)(i)		[1] shape [1] 4 numerical values Allow e.c.f.
4(c)(ii)		[1] for same amplitude and smaller maximum speed
5(c)(i)	Since copper has one conduction electron per atom, the number	